

Overview

Course Information

Class

Time: Tuesdays and Thursday, 2:00 PM - 3:20 PM

Location: Eggers 225B

Instructor

Professor Jack Reilly

Office: Eggers 225F

Office Hours: Tuesday and Thursday, 11 AM - Noon (drop in) and by appointment.¹

Zoom Hours: Friday, 1-3 ([schedule online](#))

Phone: 315-443-2687 (office)

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Course Description

Every step in policymaking relies on data. This course introduces students to data management, wrangling, communication, and visualization in the context of public policy, public administration, and behavioral science as well as the technical tools necessary to do such work in an open and reproducible fashion.

¹In addition to formal office hours, I have an open door policy - feel free to stop by if you see the door open. In particular, I'll be around for the half an hour after class each Tuesday and Thursday until 4, and my office is right next door to the classroom. Feel free to stay after and meet with me. I'll also be around after office hours until at least 12:30 each Tuesday and Thursday according to student need.

Expanded Description

Data preprocessing, wrangling, and management often consumes a large fraction of the time spent doing quantitative data analysis in public administration, public policy, and behavioral science research. Yet these topics frequently do not receive regular attention in methodological courses that focus on statistical inference. This class introduces students to the technical tools necessary to do these tasks in an open and reproducible fashion suitable for modern computational data workflows in the public sector. Throughout the course of the semester, students will learn the principles and practice of conducting reproducible quantitative research, including readable programming and coding, version control, methods of documentation, data storage, workflow management, and exploratory data visualization. A variety of relevant open technical software tools will be introduced and used, including but not limited to R (and RStudio), git (and github), markdown, and a variety of helper programs to tie things together. Special attention will be paid to data frequently used in public policy, public administration, and behavioral science.

Prerequisites

No formal pre-requisites. It is assumed you have either previously taken or are currently enrolled in an “Introduction to Statistics” or “Quantitative Methods” class (ie, PAI 721 or MAX 201), and are conversant enough in statistics to be able to work with concepts like “mean” and “standard deviation”.

While this course has no *formal* pre-requisites, it does have a substantial *informal* prerequisite: **motivation**. Learning a programming language is challenging work, and students must be prepared to invest the appropriate time, energy, effort, and - *above all* - patience.

Learning Objectives

1. Demonstrate capability in open science and contemporary reproducible data processing, wrangling, and management tools
2. Apply appropriate principles of data and file management to data projects
3. Evaluate the credibility and clarity of data visualizations
4. Create effective, reproducible, and well designed data visualizations with appropriate tools
5. Analyze large-N datasets commonly used in public policy and behavioral science

Materials

Books

- Required:
 - Weidmann, Nils. *Data Management for Social Scientists*. Open access: <https://doi.org/10.1017/9781108990424>
 - Healy, Kieran. *Data Visualization: A Practical Introduction*. Open access: <https://socviz.co>
- Recommended: A book on R programming or data wrangling
 - *Recommended*: Braun & Murdoch, *A First Course in Statistical Programming*, 3rd Edition. *Purchase links*: [cambridge](#) [amazon](#)
 - *Other options*:
 - * Freeman & Ross, *Programming Skills for Data Science*
 - * Hadley Wickham, Garrett Grolemund, and Mine Çetinkaya-Rundel, *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*, 2nd ed. <https://r4ds.hadley.nz>.

i A Programming Book

There are, in essence, three kinds of books that are useful for the class - a book on data management, a book on data visualization, and a book on data programming. For the first two, I've required the open access books by Weidmann and Healy (**DMSS** and **DV**, respectively). For the third, you have options. I recommend Braun and Murdoch (**FCSP**), which is a good general overview of the R language from a statistical programming perspective. Freeman & Ross (**PSDS**) is a more general introduction to the overall data science technical environment, and Wickham et al (**RDS**) has the advantage of being open access (always useful). **PSDS**, however, doesn't have as detailed information on actual programming in it, and **RDW** but is primarily focused on the tidyverse, which our course is not exclusively focused on. We will cover all programming and scripting content in class, so the book you choose for background reference is up to you, but you will find it valuable to have one.

Computing

You will need access to a personal computer for this class. It will need to run a full operating system, where you have the ability to install local applications outside of app stores and have access to the command line. MacOS, Windows, and Linux are all fine. Tablet or web-book OSes - like Chromebooks or iPads - won't be sufficient. Aside from the computer, all significant software we use will be free/open-source, and we'll cover usage and installation in class.

Online Course Resources

Blackboard is our internet-based course platform: <http://blackboard.syr.edu>. In it, you will find submission portals for assignments and a link to the [course webpage](#), where you can find the course syllabus, problem sets, and links to readings.

You can also link to our [course drive here](#), which contains lecture slides, data sets, and some other useful things for the class.

Please note that class attendance is the primary source of course-related announcements and material.

Course Requirements

Overview

Satisfactory completion of the course requires completion of the following:

- Regular course participation and attendance (10%)
- Weekly Assignments (30%)
- Practicum (Take Home) (15%)
- Core Exam (In Class) (15%)
- Final Project (30%)

Attendance

One of the guiding principles of my class is that you are adults, and thus, capable of managing your own time. I have little interest in policing your lives. Attendance is kept for each day of class, but you will lose no points on attendance if you happen to miss a couple days: everyone has things that occasionally come up in life that need to be dealt with, and I fully realize that some of those things are things you - very understandably - may not want to discuss with your professor. *That's OK!*

That said, attendance in class is an important element to doing well in the course. If you must miss more than a couple days, it's a good idea to check in with me so that I don't mark you off for chronic absenteeism. The easiest way to do this is just email me with a brief reason when something comes up and you have to miss class (which will also allow me to tell you if you're missing anything particularly important).

If you must miss class, the way to make up what you've missed is straightforward: make sure to look over the posted material, do the reading, get notes from a friend, and still complete the assignment if you are able (and make sure to look over any assignment solutions). If you

do these things and still feel like you're missing something, please feel free to come into my office hours and we can talk it through.

i Participation

There is no formal grade for “participation”. However, I reserve the right to dock a couple points here if you do ridiculous/unprofessional things in class (like answering your cell phone, always coming in late and regularly distracting others, spontaneously breaking out into ribald song in the middle of class, etc).

Assignments

There is an assignment each week in class, **due Thursday by class time**.

Assignments will vary in nature: some will be one-off problem sets, some may build on problem sets from a prior week. All material needed for an assignment will be covered by the Tuesday before the assignment is due (usually much earlier), and the assignment itself will be given a week ahead of time. **No assignment work is accepted after class**, as we will go over answers for assignments in class.

Students may miss up to two assignments with no penalty. Students may also work together on assignments - in fact, I encourage you to do so - although each student is ultimately responsible for their own learning and work.

i Effort-Based Evaluation

Assignments are evaluated based upon effort and a check completion system. Students who answer every problem will earn a check, with each check worth one point toward their final assignment grade.

Practicum

A practicum is a large assignment that is worth more and graded on a scale.² It is untimed, take-home, cumulative, and will be completed on your own time (and computer). Unlike the weekly assignments, you are not allowed to work together on the practicum. Essentially, think of it as take home test that complements the in-class core exam.

²While regular weekly assignments are evaluated on an effort based, check/no-check system, on the practicum, you'll receive a certain number of points out of the total, like a test.

Core Exam

The core exam will be in class. More information will be given as the exam gets closer.

Final Project

A project utilizing data of your own choice. Graduate students will have higher expectations than undergraduate students. More information will be given as the exam gets closer.

Course Expectations & Guidelines

Etiquette & Decorum

This is a graduate course: I take it for granted that you have a basic interest in the material, an enthusiastic attitude toward participation, and a respectful attitude to everyone in the room. A university classroom is fundamentally a learning community: be courteous to fellow students and the professor, don't let yourself be distracted by your cell phone in class, and don't let what is on your computer screen distract fellow students in the class, either.

Office & Consultation Hours, Appointments

I encourage you to chat with me at any point if you have questions about the course. You can schedule a meeting with me by going to my website here: <http://jackkreilly.github.io> and sign up for time at your convenience. You can also always just drop in during my regularly scheduled drop-in office hours without appointment or stop by to see if my door is open: if it's open, come on in. (Don't feel like you're intruding! I'll tell you if it's not a good time.)

E-mail

Email is the best way to contact me. I'm usually pretty responsive, but as a baseline, I always aim to get back to you in a modified 24-hour fashion: by the end of the business day the day after you email. So if you email me at 2 PM Tuesday, I'll get back to you by 6 PM Wednesday at the latest; if 10 PM Thursday, by 6 PM Friday; if you email me at 3 PM on Friday, by 6 PM Monday, etc.³

³Again: usually I'm much faster! But if you don't hear from me by this baseline, feel free to bump a reminder.

i Note

If your email requires a long response, expect me to encourage you to schedule an appointment with me so that we can more effectively discuss the matter.